



AI NATIONAL EMERGENCY RESOURCE BRAIN

A centralized AI coordination layer that predicts, allocates and routes emergency resources across India's NDMA – SDMA – DDMA disaster response network in real time

Predictive AI • Disaster Resource Allocation • Public Sector Technology



INNOVEXA

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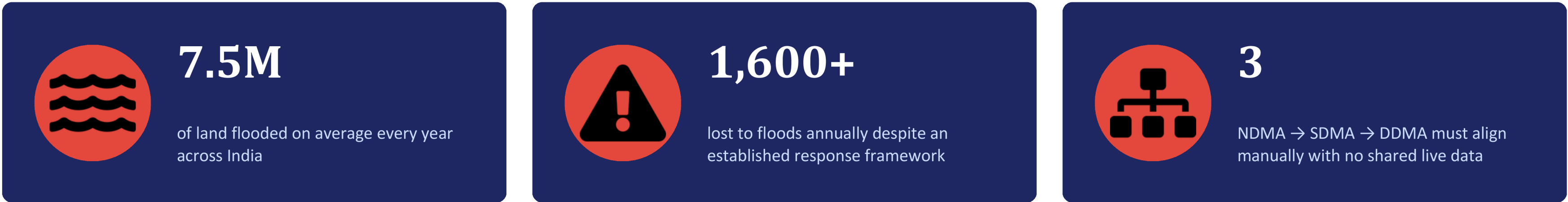
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Problem Statement

India's emergency response runs on a fragmented, reactive system — even with strong policy on paper

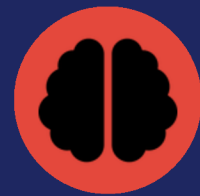


Source: Government flood-damage and disaster-governance data (NDMA / EVS Institute) — figures shown as approximate national averages



Our Solution

One AI coordination layer connecting every agency, sensor and resource in the national network



AI National Emergency Resource Brain sits above NDMA, SDMA and DDMA as a shared intelligence layer — continuously sensing demand, forecasting resource needs with machine learning, and automatically allocating the nearest available assets to the point of crisis.



Predict

LSTM / Prophet forecasting models flag where and when demand will spike, using weather, seismic and historical data.



Allocate

A constraint-based optimization engine assigns the nearest, most suitable resource to every request in seconds.



Coordinate

One shared dashboard aligns NDMA, SDMA, DDMA, hospitals and volunteers on a single live response plan.

Where It Fits Nationally

The AI Brain overlays India's existing three-tier disaster management hierarchy — it doesn't replace it



Key Features

Core capabilities that make coordination fast and data-driven



Real-Time Resource Tracking

Live GPS status of ambulances, fire units, shelters and medical stock nationwide.



Predictive Demand Forecasting

LSTM / Prophet models anticipate surges from weather, seismic and health-trend data.



Automated Allocation & Routing

Graph-based optimization dispatches the nearest suitable resource in seconds.



Multi-Agency Dashboard

One command view spanning NDMA, SDMA, DDMA, hospitals and municipal teams.



Smart Alerts & Escalation

Automatic priority alerts the moment demand starts to outpace available supply.

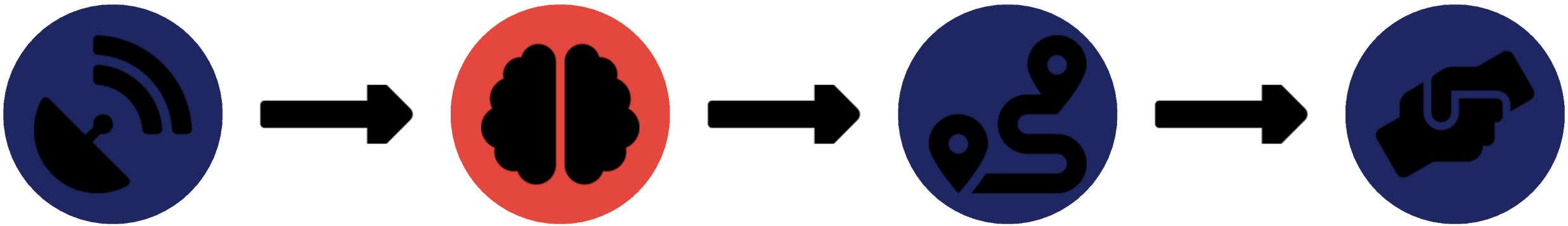


Citizen Reporting App

Crowdsourced, geo-tagged incident reports feed the AI engine in real time.

System Architecture

From raw signals to on-ground action, with a feedback loop that keeps improving both



Data Sources

AI Engine

Decision Layer

Field Action

Weather (IMD), seismic feeds, hospital APIs, GPS fleets, citizen reports

LSTM / Prophet forecasting + graph-based route optimization

Priority scoring, best-fit match, route & ETA computed

NDMA / SDMA / DDMA teams act on one live dashboard



Forecasting: LSTM & Prophet time-series models



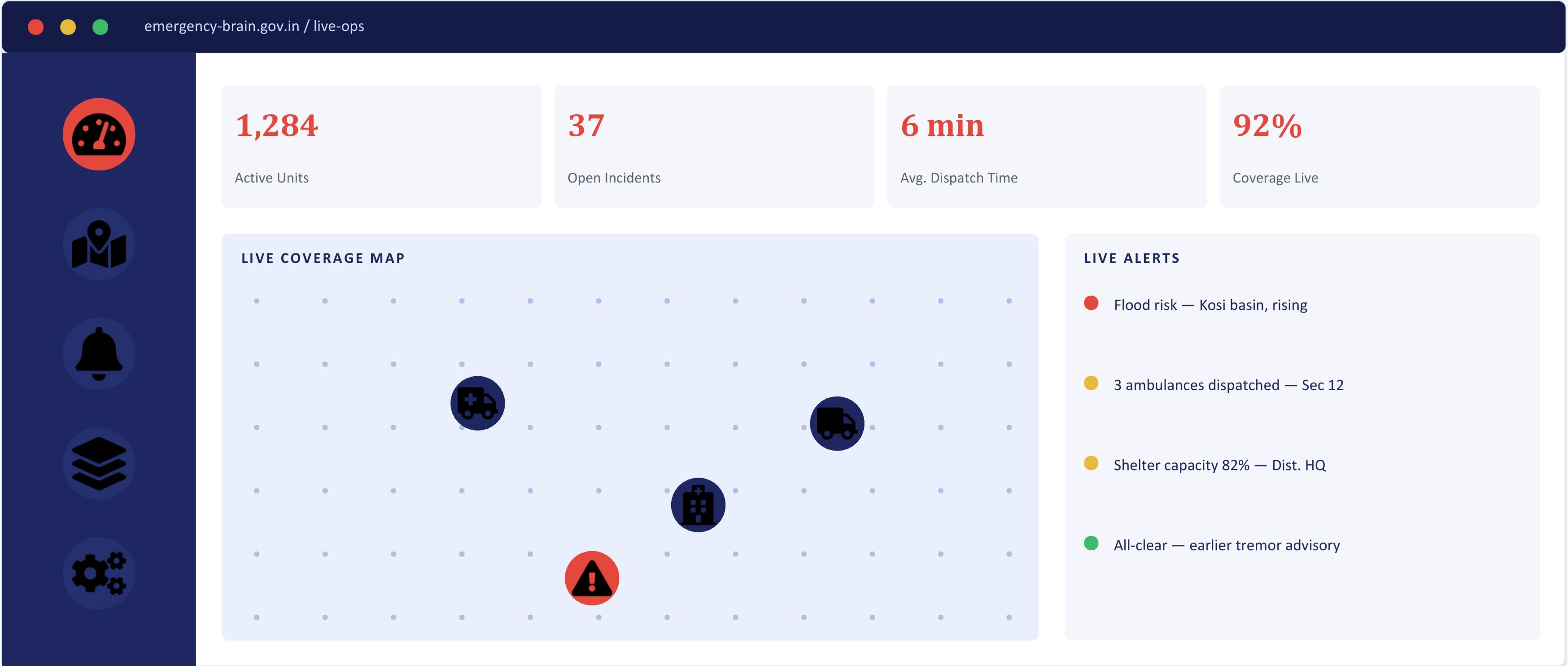
Allocation: constraint-based optimization



Routing: Dijkstra / A* over live road-network graphs

Inside the Dashboard

A single live operating picture shared across every responding agency



The Brain in Action

A flood scenario, from first signal to boots on the ground — in under ten minutes



Technology Stack

Built with a modern, scalable, cloud-native stack



AI / ML

Python · TensorFlow · Prophet · scikit-learn · Pandas



Backend

Node.js · REST APIs · WebSockets



Frontend

React · Tailwind CSS · Mapbox GL



Data & Storage

PostgreSQL · PostGIS · Redis



Cloud & DevOps

AWS / GCP · Docker · Kubernetes



Security

OAuth2 · Role-Based Access · Encryption

Expected Impact

The measurable difference a shared, predictive coordination layer can make



40%

Faster average
dispatch time



3x

Improvement in
resource utilization



24/7

Continuous
national monitoring



1

Shared dashboard for
NDMA, SDMA & DDMA

Targets are illustrative — based on optimization-model simulations, not a live deployment.



Ultimately, this means saved lives. By closing the gap between a crisis and the nearest available help, the system turns hours of manual coordination into seconds of data-driven decisions — across floods, earthquakes, fires and public-health emergencies alike.

Future Scope

Where the AI National Emergency Resource Brain goes next



Drone-Based Damage Assessment

Autonomous aerial survey of disaster zones to feed real-time visual data into the AI engine.



IoT Sensor Integration

Direct feeds from flood gauges, seismic sensors and smart city infrastructure nationwide.



Multilingual Citizen Access

Voice and chat-based reporting in regional languages for wider, more inclusive reach.



Expanded Disaster Coverage

Extend models beyond floods and quakes to wildfires, industrial accidents and pandemics.



Thank You

AI National Emergency Resource Brain — one AI coordination layer for a faster, smarter, more connected national emergency response.